

VIROLAHTI KOIVUNIEMI, Finland

WMO: 028310

Lat: 60.527N

Lon: 27.673E

Elev: 16

StdP: 14.69

Time Zone: 2.00 (EUE)

Period: 08-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-11.6	-5.1	-16.3	2.3	-11.4	-9.2	3.4	-5.0	20.3	34.5	18.7	33.4	2.6	0	0.619

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	17.7	80.3	67.2	76.8	65.9	73.5	63.6	69.9	77.2	67.5	74.5	65.3	71.6	7.1	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
67.1	99.6	74.3	64.7	91.6	71.6	62.6	85.1	68.9	33.9	77.2	32.0	74.4	30.2	72.0	75.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4)	18.7	16.6	14.8	DB	-13.7	82.0	10.5	4.8	-21.3	85.4	-27.4	88.3	-33.3	91.0	-40.9	94.5
(5)				WB	-14.1	71.3	10.4	2.5	-21.6	73.1	-27.7	74.5	-33.5	76.0	-41.1	77.8

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	40.9	19.6	22.2	27.5	37.5	49.1	56.3	61.9	59.6	51.8	41.0	35.0	27.8
	DBStd	16.27	12.51	11.39	8.00	5.70	7.52	5.83	5.80	5.21	5.63	6.42	7.47	9.99
	HDD50	4388	943	779	697	378	108	13	1	3	44	284	449	689
	HDD65	8863	1408	1199	1162	826	493	268	134	180	395	745	899	1154
	CDD50	1056	0	0	0	2	81	201	368	301	99	4	0	0
	CDD65	58	0	0	0	0	1	6	37	14	0	0	0	0
	CDH74	631	0	0	0	0	29	96	374	132	0	0	0	0
	CDH80	93	0	0	0	0	1	10	66	16	0	0	0	0
Wind	WSAvg	7.0	6.9	6.7	7.1	6.3	6.9	7.1	6.4	6.5	6.7	7.5	7.9	8.3
Precipitation	PrecAvg	27.0	2.2	1.9	1.6	1.3	1.6	2.2	2.3	3.0	2.6	3.0	2.8	2.4
	PrecMax	37.5	4.3	4.4	3.4	2.9	4.3	5.3	5.1	6.8	6.0	7.4	6.6	5.7
	PrecMin	19.7	0.2	0.0	0.1	0.2	0.4	0.3	0.1	0.2	0.6	0.8	0.3	0.5
	PrecStd	4.0	0.9	1.1	0.8	0.7	1.1	1.0	1.2	1.5	1.7	1.7	1.3	1.1
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	39.1	40.8	45.6	64.8	77.5	81.1	85.3	82.1	69.9	56.7	50.3	45.7
		MCWB	37.9	36.7	38.3	51.4	61.2	65.5	69.9	68.4	62.3	52.9	48.7	45.1
	2%	DB	37.6	37.4	42.1	57.7	73.2	76.6	81.7	77.7	66.3	54.6	48.0	42.9
		MCWB	36.7	35.4	37.1	47.3	59.0	64.4	69.0	67.2	60.1	51.7	46.7	41.7
	5%	DB	35.9	35.6	39.2	52.1	69.1	72.7	78.4	73.7	64.0	53.1	45.9	40.8
		MCWB	35.4	34.6	35.7	44.3	56.5	61.0	67.8	64.5	58.7	50.8	44.8	39.7
	10%	DB	34.0	34.4	36.8	48.7	64.9	69.1	74.6	70.6	62.0	51.3	44.3	38.9
		MCWB	33.6	33.6	34.5	42.1	54.3	59.7	65.4	63.2	57.7	49.2	43.3	37.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	38.4	37.4	40.1	52.2	64.8	69.0	73.2	71.7	63.8	54.5	49.2	45.2
		MCDB	38.8	39.6	42.9	61.2	73.9	77.2	81.1	79.7	67.9	56.0	50.0	45.4
	2%	WB	36.9	35.9	37.9	48.6	61.0	65.8	70.8	68.6	61.5	52.9	46.8	42.0
		MCDB	37.4	37.0	40.5	56.0	68.9	73.9	78.9	74.7	64.9	54.0	47.7	42.6
	5%	WB	35.5	34.8	36.2	45.5	58.1	63.0	69.0	66.1	60.0	51.4	44.9	39.8
		MCDB	35.9	35.6	38.2	50.9	66.6	70.3	76.3	72.4	62.8	52.6	45.7	40.7
	10%	WB	33.7	33.7	34.9	43.0	55.1	60.7	66.7	64.1	58.5	49.4	43.5	37.8
		MCDB	34.0	34.3	36.5	47.3	63.5	67.4	73.4	68.9	61.0	51.0	44.2	38.6
Mean Daily Temperature Range	5% DB	MDBR	10.9	10.5	13.2	16.3	20.2	17.8	17.7	17.7	14.9	11.3	7.3	8.1
		MCDBR	5.7	7.5	13.9	24.6	26.4	24.9	23.3	23.5	18.7	13.6	8.0	7.6
		MCWBR	5.4	6.8	10.2	15.7	14.4	13.4	12.5	13.9	13.4	11.2	8.0	7.8
	5% WB	MCDBR	5.6	7.3	11.1	21.7	23.2	21.3	20.2	19.6	15.2	11.5	8.0	7.4
		MCWBR	5.4	6.6	8.6	14.7	13.6	13.0	11.7	12.5	11.6	10.3	7.9	7.6
Clear-Sky Solar Irradiance	taub	0.275	0.310	0.332	0.370	0.367	0.365	0.388	0.378	0.353	0.333	0.302	0.381	
	taud	2.122	2.150	2.200	2.284	2.408	2.457	2.412	2.433	2.488	2.383	2.198	2.591	
	Ebn at Noon	146	210	248	261	272	275	264	257	245	205	135	101	
	Edh at Noon	13	23	31	34	33	32	33	30	24	19	13	9	
All-Sky Solar Radiation	RadAvg	80	272	745	1221	1669	1741	1738	1327	786	324	87	42	
	RadStd	18	58	91	131	182	113	157	138	90	48	15	5	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (45 neighbors)	+0.94	N/A	N/A	N/A	N/A	N/A	-315	-340	N/A	N/A

Nomenclature: See separate page